STUDENT SKILLS AND COURSE RECOMMENDATION SYSTEM

Research Paper 1: A Skill-Gap Analysis Approach for Personalized Course Selection

Student Research Paper | October 2026

Abstract

Choosing a course can be difficult when students do not clearly understand their strengths, interests, or the skills needed for a future career. This paper proposes a student skills and course recommendation system that identifies a learner's current competencies, compares them with the skills required for a selected goal, and suggests suitable courses. The proposed approach uses a structured student profile, a course-skill database, and a transparent matching score. It is a conceptual design rather than a report of a completed experiment. The paper discusses the workflow, possible benefits, limitations, and ways to evaluate the system.

Keywords: student skills, skill gap, course recommendation, personalized learning, career planning

1. Introduction

Students often select courses because they are popular, recommended by friends, or widely advertised. However, a course that suits one student may not suit another. Students have different levels of knowledge, interests, learning preferences, and career goals. A recommendation system can help organize this information and present options that are relevant to each learner. The aim is not to make a final decision for the student, but to support an informed choice.

2. Problem Statement

A general course list does not show how closely each course matches a student's existing skills or future plans. Students may also find it difficult to identify the gap between what they can do now and what a target course or career expects. The proposed system addresses this problem by comparing student skills with course requirements and explaining why a course is recommended.

3. Objectives

The main objectives are: (i) to create a profile of a student's skills and interests; (ii) to record the skills developed by different courses; (iii) to identify missing or weak competencies for a chosen goal; (iv) to rank courses according to their relevance; and (v) to provide understandable reasons for each recommendation.

4. Proposed Methodology

The system can work in five stages. First, the student enters interests, completed subjects or courses, self-assessed skills, and a career or learning goal. Second, the system stores course information, including prerequisites, topics, difficulty level, and skills developed. Third, each skill is represented using a common scale, such as beginner, developing, proficient, or advanced. Fourth, the system compares the student profile with course requirements. Finally, it displays a ranked list of courses with short explanations.

5. Skill-Gap Matching Model

For each skill, the system can compare the required level with the student's current level. A positive difference indicates a possible learning gap. Courses that teach the missing skills receive a higher relevance score. A simple

model could assign weights to important skills and calculate how much each course supports the selected goal. The weights should be visible and adjustable rather than hidden from the student. The score should be treated as guidance, not as an objective measure of a person's ability.

6. Expected Benefits

The approach may help students discover relevant courses, avoid choosing only on popularity, and plan learning in smaller steps. It can also help teachers or counsellors discuss skill development with students. Showing the reason for a recommendation can make the system easier to understand and allow students to reject suggestions that do not fit their interests.

7. Limitations and Ethical Considerations

Self-assessments can be inaccurate, and course descriptions may be incomplete or outdated. A student's interests and goals may change over time. The system should therefore allow profiles to be updated and should avoid treating a low score as a permanent judgement. Student data should be collected only when necessary, stored securely, and used with appropriate consent. Recommendations should be checked for unfair patterns and should not replace guidance from teachers or counsellors.

8. Evaluation Plan

A prototype could be evaluated with students and teachers. Useful measures include how relevant students find the recommended courses, whether explanations are understandable, how often students select a suggested option, and whether recommendations cover a range of suitable choices. The system could also be compared with a simple search-by-category method. Any findings should be reported with the sample size and study limitations; no performance results are claimed in this conceptual paper.

9. Conclusion

A skill-gap approach offers a practical foundation for a student skills and course recommendation system. By comparing current competencies with the skills taught by courses, the system can suggest useful next steps and explain its choices. Its success would depend on reliable course data, careful evaluation, privacy protection, and keeping the student in control of the final decision.

References

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• These references provide related background; the proposed model and evaluation plan in this paper are a conceptual design.